THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION - TERMINAL EXAMINATION
BIOLOGY FORM THREE - MAY - 2021
MARKING SCHEME

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
A	A	D	A	D	A	B	C	D	D

(10 marks).

2.

i	ii	iii	iv	v
C	D	B	E	A

(5 marks).

3 (a). A person with blood group AB is not always considered a universal recieipient because of rhesus factor, that is a person with AB can either be AB^+ or AB^-. This indicate that a person with positive rhesus factors can not give to a negative rhesus blood group. (2 marks)

b(i) Blood transfusion refers to the transfer of blood from one person to another through blood vessels (1 mark).

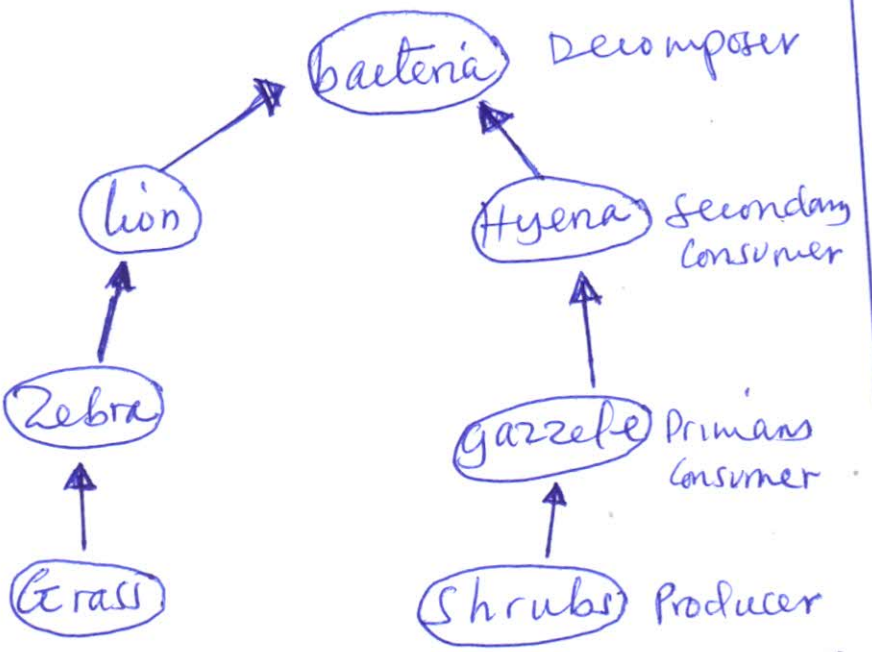
(i). The blood of donor should be investigated
(ii). Blood groups should be established
(iii). Donor should be healthy and aged 18-60
(iv) Donor should be given biscuits or glucose (Any 3 1mark@ = 3 marks)

4 (a) (i) Ecology refers to the study of organisms in relation to their environment. (1 mark).

(ii) Environment refers to the totality of all things that surrounds us both biotic and abiotic. (1 mark).

(iii) Ecosystem refers to the natural unit composed of both biotic and abiotic in which they interact each other. (1 mark)

b



(3 marks)

5 (a). Disease is an abnormal conditions that interferes with the health of the body of an organism (1 mark).

b (i). Malaria 1 mark

(ii). Caused by plasmodium
- Transmitted by vector female Anopheles mosquito
1 mark

b(iii) Can be prevented Through
- sleeping under treated nets
- Draining stagnant water
- Using Mosquito repellants
- Cutting long grasses
- Using anti-malaria drugs

Any 1@1 = 1 mark

(c)

Disease	Causative agent	Vector
Malaria	Plasmodium	female Anopheles mosquito
Cholera	Vibrio Cholerae	Housefly.

4@½ = 02 marks

6 @ (i). Producing tears which kill microorganisms
(ii). Releasing sweats which remove bacteria away
(iii) By sneezing which free bacteria out of the body
(iv). Secretion of the Hydrochloric acid kill bacteria

4@1=04marks

(b). (i) Because during the night plants consume Oxygen gas therefore there would be competition for O_2 gas which can lead to some Complications like fainting, etc

(ii). Because Charcoal burner produces more Carbon monoxide and Carbondioxide which is harmful to man

2@1=02 marks.

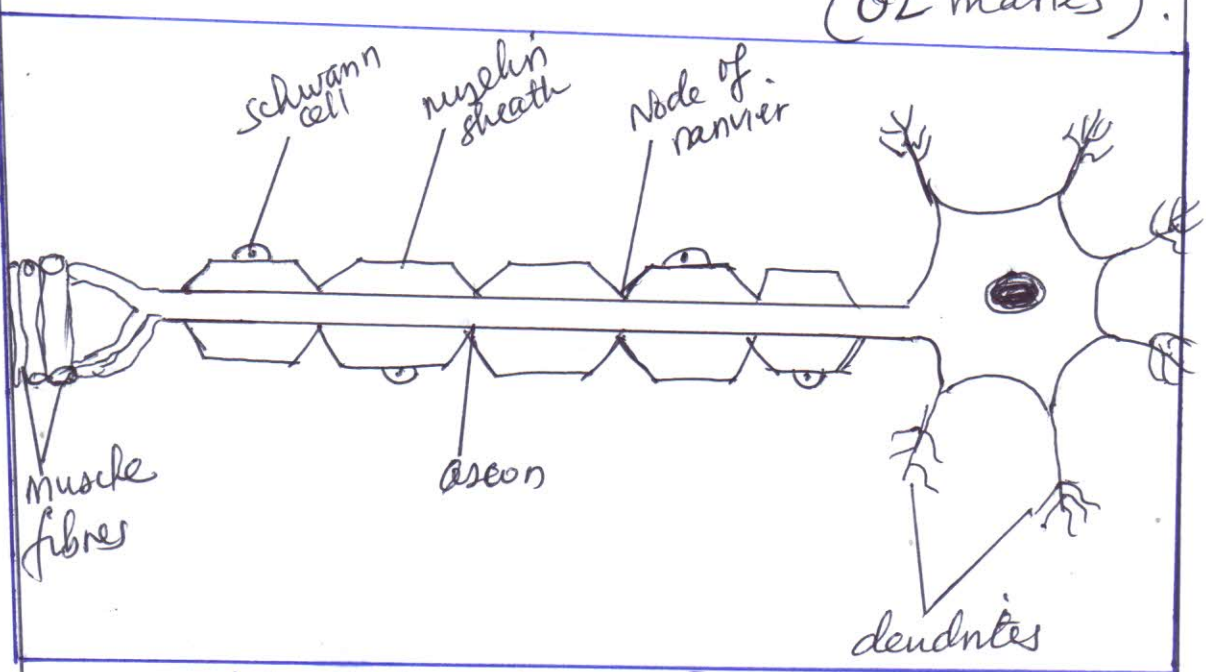
7 @ Four (4) ways to help a snake bite
- Lay a victim down
- Tie above bitten region to slow blood flow
- If breathing is difficult help mouth to mouth
- Take a patient to a hospital.

4@1 = 04 marks.

(b) - Do not give alcohol
- Avoid any act that can speed rate of heartbeat

(02 marks).

8 (a)



MOTOR NEURONE

3 marks

(b)
- Motor neurone
- Relay/intermediate neurone
- Sensory neurone

3@½ = 1½ marks

(c)

Nervous communication	Hormonal Communication
(i) involve nerves	Involve glands
(ii). It is very fast	It is slow
(iii) Response is short lasting	Response is long lasting
iv). Parts affected are effectors	Parts affected are internal organs such as liver, glands

3@½ = 1½ marks

9 @ Excretion refers to the process in which waste products resulted from cell metabolism are removed out the body. (1 mark)

(b) - Salt
- Excess water
- Urea
- Ammonia Any 3 @ ½ = 1½ marks

(c). Urine formation involves three processes which occur in the nephrone.
- Ultrafiltration ½ mark
- Selective reabsorption
- Secretion

Ultrafiltration: takes place in glomerulus whereby substances are filtered where some pass like amino acids, water, glucose, ions, urea and some large substances like ~~blood cell~~ proteins are retained
1 mark

Selective reabsorption: occur in proximal tubule and loop of Henle where some useful materials are reabsorbed back into the blood stream like water, glucose, ions, etc.
1 mark

Secretion: Some wastes are secreted to the tubule in collecting duct like drugs and toxin where they join to form liquid called urine which then collected by duct and passed to the urinary bladder through ureter.
1 mark

10 (a) Photosynthesis refers to the process in which all green plants and green algae manufacture own food using light from the sun and some inorganic substances like carbondioxide and water. (1 mark).

b) - Light from the sun
- Chlorophyll
- Water
- Carbondioxide 4 @ ½ = 02marks

(c). $6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6H_2O + 6O_2$

(03 marks).

11 @ (i) Tropism is a growth movement of plant in response to a certain stimulus. (1 mark).

(ii). Hydrotropism is growth movement of plants towards a source of water. (1 mark)

(iii). Phototropism refers to the growth response of plants towards a source of light. (1 mark).

(b). Plants bent towards a source of light due to concentration of Auxins hormones which are secreted more in dark region to make plant grow faster in dark region and hence the plant bent towards a source of light. 3 marks

12 (i) Muscles refers to specialized tissues for contraction and relaxation to bring about movement. (1 mark).

(ii). Muscle cramps refers to the involuntary contractions of muscles due to the fatigue. (1 mark).

(b) Possible causes of Muscle cramps.
- Dehydration
- Lack of salt.
- Muscle fatigue.
- Vigorous excercises

Any 3@1 = 03 marks

(c) - regular physical excercise
- take some drinks
- Take salts

Any 2@½ = 01 mark

SECTION C 25 marks

13 Drugs - refers to the chemical when taken bring a lot of problems (effects) to man 1 mark

Drug abuse refers to the illegal use of drugs. 1 mark

Causes of drug abuse

- Peer pressure
- Sickness
- Avoid some realities of life
- Reduce stress

(04 marks)

Effects of drug abuse

- liver damage
- Cancer
- Street children
- Brain damage
- Increase of Crime

(04 marks)

Prevention of drug abuse

- Provision of Education
- Engage in different works
- Avoid peer groups
- Provision of employment

(04 marks).

Conclusion - (01 mark).

Introduction = ½ mark

14 (a) Components of blood

- Plasma
- White blood cells
- Red blood cells
- Platelets

4@2 = 08 marks

(b) Functions of blood

- Transport of nutrients
- Transport of excretory waste materials
- Transport of hormones
- Transport of gases
- Regulate body pH and temperature

Any 3@2 = 6 marks

Conclusion : 05 mark

15 Balanced diet refers to a diet that contains all types of foods in a right proportions 1 mark

Components of balanced diet

- Carbohydrates
- Proteins
- Vitamins
- Lipids
- Roughages
- Water (2 marks)

Importance of balanced diet

- Provide body energy
- for growth
- For metabolic processes (2marks)

Deficiency disorder of balanced diet

- Kwashiorkor
- Night blindness
- Marasmus
- Goitre (2 marks)

Two factors that contribute to the lack of balanced among children in a community.

- Ignorancy
- poverty (2marks)

Conclusion: (1 mark)